

Tom Derrick

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PROFILE

I'm a data scientist with four years across forecasting, mathematical optimisation, and applied AI — currently leading the development behind Modo Energy's Australian products. My work runs the full stack: MILP models solving over real transmission networks, ML price forecasters, the AWS pipelines that run them, and the React frontends customers touch. I direct our NEM model roadmap, reporting to the Head of Region, and train and delegate to a junior data scientist.

EXPERIENCE

Data Scientist — Modo Energy, Sydney

Aug 2025 – present

Series B scaleup building AI-driven forecasting and investment intelligence for the energy transition — customers include Goldman Sachs, Apollo, EDF, and Shell. I joined the Australian team in its first year and lead the modelling behind its local products.

- Designed and shipped the nodal transmission network model behind the July 2026 forecast release — DC optimal power flow, bus-level dispatch, PTDF, load disaggregation, transmission augmentation through 2050+. Took the product from regional to 200+ nodes: largest update since the Australian launch and its first on-time release, delivered as sole modeller while training a new data scientist.
- Designed the capacity expansion model (linopy + XPRESS MILP) that replaced weeks of manual capacity-buildout tuning per forecast with a single optimisation at minimal compute cost.
- Build and run the production cost models producing long-term NEM price forecasts; battery dispatch optimisation and revenue modelling in the multi-market Global Dispatch Model (Pyomo + XPRESS).
- Built the full-stack contracting strategy system — backend valuation engine plus React configuration UX shipped to customers.
- Built and deployed an LLM support agent (Claude + RAG over model docs, code, and outputs) answering customer and internal questions across Slack and Intercom; contributed data engineering and semantic table annotation to Modo's AI assistant.
- Authored Modo's public "Australia NEM Forecast: July 2026 release" and "NEM Market Outlook Report: Q2 2026" research.

Market Analyst (Graduate, Jul 2022 – Feb 2023) — Delta Electricity, Sydney

Jul 2022 – Jul 2025

- Designed whole-of-market NEM simulation models (PLEXOS/Prophet) used for trading, hedging, and business development.
- Built an ML-powered spot-price forecasting model and a Monte-Carlo Gross-Margin-at-Risk engine for risk compliance and portfolio optimisation.
- Modelled major investment decisions — including the end-of-life extension of Vales Point Power Station — and LGC/ACCU certificate trading strategy.
- Optimised the planned-outage schedule to minimise revenue impact; built the internal live trading-performance dashboard. Completed Delta's 2024 future-leadership program.

Customer Success Manager — TechTarget, Sydney

Nov 2020 – Jun 2022

- Managed APAC campaign delivery for marquee accounts (Red Hat, Dell, IBM, Cisco, Alibaba).

PERSONAL PROJECTS

- **NemDash** (github.com/tder311/NemDash) — real-time electricity-market analytics platform: 5-minute AEMO ingestion pipeline, XGBoost price forecaster, LP battery dispatch optimiser, bid-band generation, tool-calling LLM chat agent. FastAPI · PostgreSQL · React.
- **opencode-orchestrator** (github.com/tder311/opencode-orchestrator · npm) — opencode plugin for parallel AI coding agents: each spawns into an isolated git worktree with its own interactive terminal session, pushing completion/blocker reports back to a supervising session.
- **orchestrator** — coordinates multiple AI coding agents working in parallel across codebases: worktree isolation, task routing, session management.

EDUCATION

- UNSW — Master of Data Science (with Excellence)
- University of Sydney — BSc, Computer Science & Statistics
- AFMA Financial Markets Accreditation

SKILLS

Python · linopy · Pyomo · XPRESS · XGBoost · AWS (Batch, S3) · FastAPI · React · PostgreSQL · Claude API